SRN

PES University, Bangalore

(Established under Karnataka Act 16 of 2013)

UE17CS351

END SEMESTER ASSESSMENT (ESA) B. TECH. 6th SEMESTER – DEC-2020

UE17CS351 - Compiler Design

Time: 3 Hrs | Answer All Questions | Max Marks: 100

1.	a)	Explain the different phases of the compiler using following c code: **While(a<b && b<c) {a=10 ; } c=a+b;**	10
	b)	Write the transition diagram to recognize the relation and logical operators	8
	c)	With block diagram explain the components of lex	2
2.	a)	With block diagram explain the components of lex? Explain the buffer pairs approach used for the input buffering technique.	8
	b)	Construct SLR(1) parsing table for the grammar given below . S→ CC C → cC|d State weather the above mentioned grammar is in LR (0) and SLR(1)	12
3.	a)	Write SDD to compute the types and width T → B C B → int B → float C → [num] C	8
	b)	Using the SDD of Q- 3A draw the dependency graph for the string array (2, array(3,integer))	8
	c)	Along with examples explain the different types of attributes used to define the semantic rules	4
4.	a)	Translate the following code into 1. Quadruples 2. Triples 3. Indirect Triples If(a<b) c=func(a,b); printf("hello");	10
	b)	Mention the different types of issues involved in code generation?	8
	c)	Explain the subdivision of runtime memory	2
5.	a)	How to access the non local data on the stack	5
	b)	With example explain the Control flow graph and basic blocks	10
	c)	Mention the general format of activation record.	5